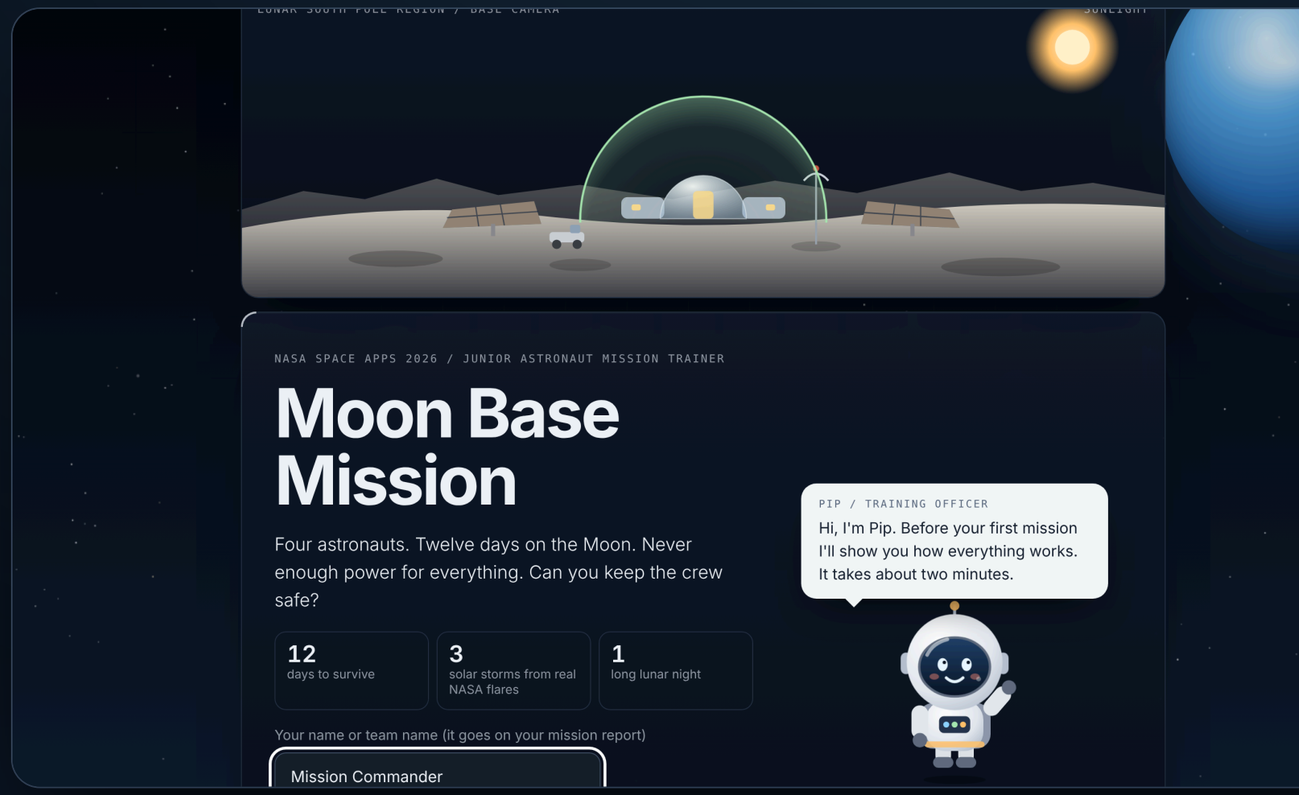
NASA SPACE APPS 2026 / JUNIOR ASTRONAUT MISSION TRAINER

Moon Base Mission

A browser game that trains young astronauts to keep a Moon base alive for 12 days, with solar storms built from **real NASA flare data**.

itzzmeeneha.github.io/moon-base-mission



Code and text written with help from Claude (Anthropic), an AI assistant

THE CHALLENGE

Space lessons are often just facts. We wanted kids to feel an astronaut's hard choices.

You run a Moon base with four astronauts. Every day the solar panels give a few power cells, and you decide where they go: **air**, a **radiation shield**, or **science**. There is never enough for everything.

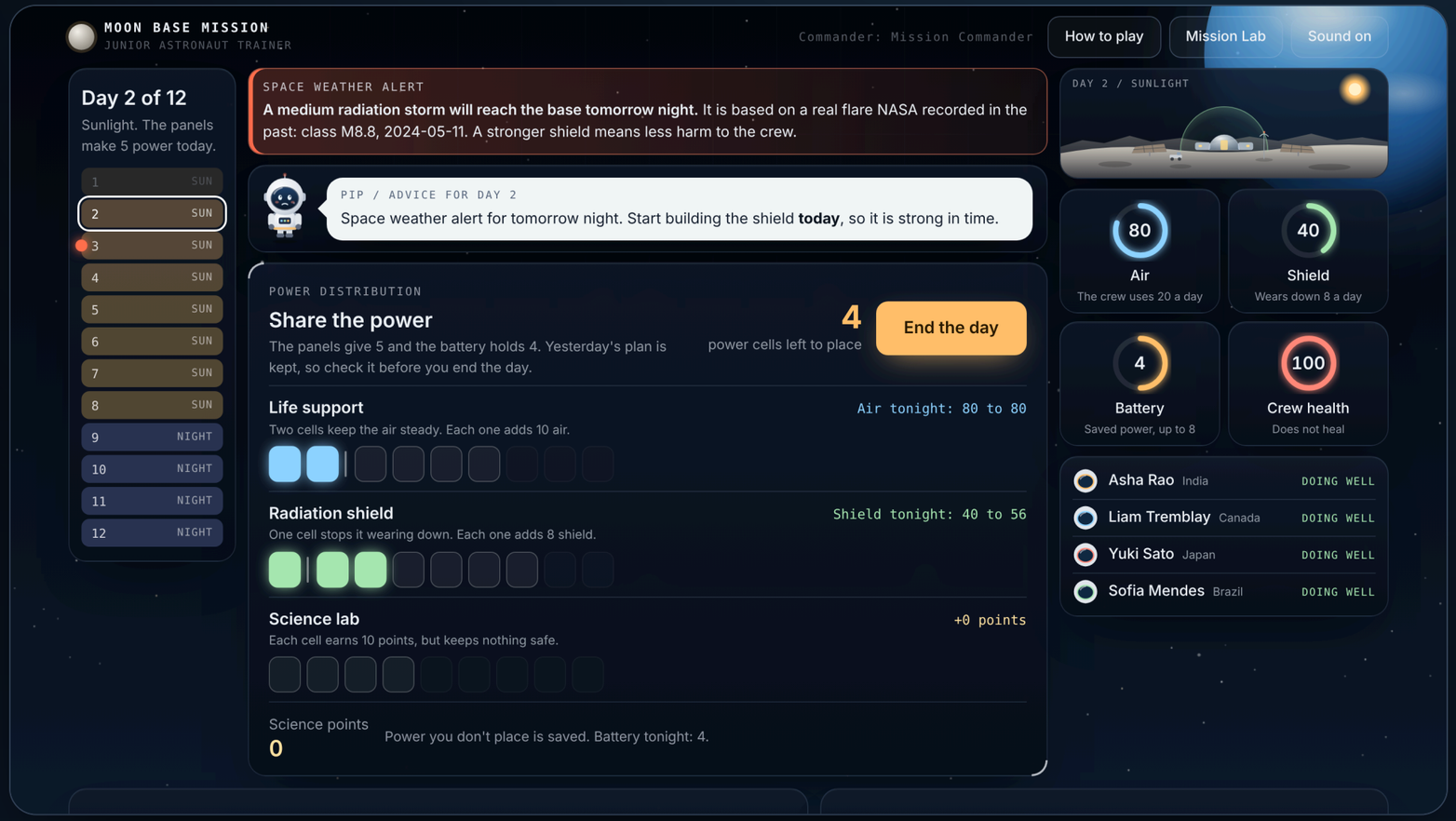
12	3	3	1
days to survive	choices every day	solar storms from real NASA flares	long lunar night

One HTML file. Works in any browser, on phones and school computers, with no install or account.

HOW IT PLAYS

Every choice shows its result tonight

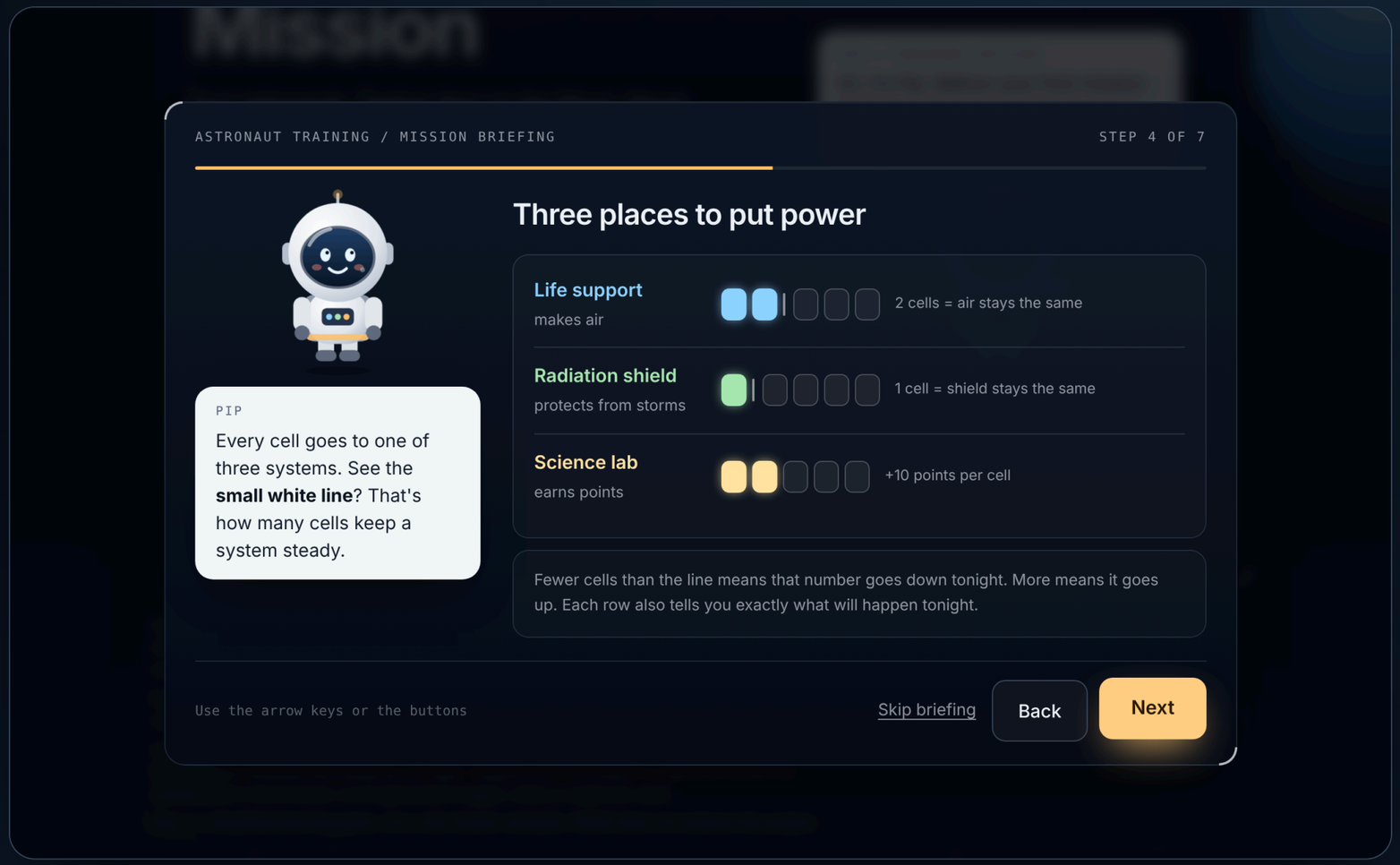
- Tap boxes to place power cells. Each row says what happens tonight.
- The **lunar night** cuts solar power, so players save energy in a battery.
- A **space weather alert** warns of each storm a day early.
- The whole mission fits on one screen.



BEGINNER FRIENDLY

Pip, the guide, trains every player

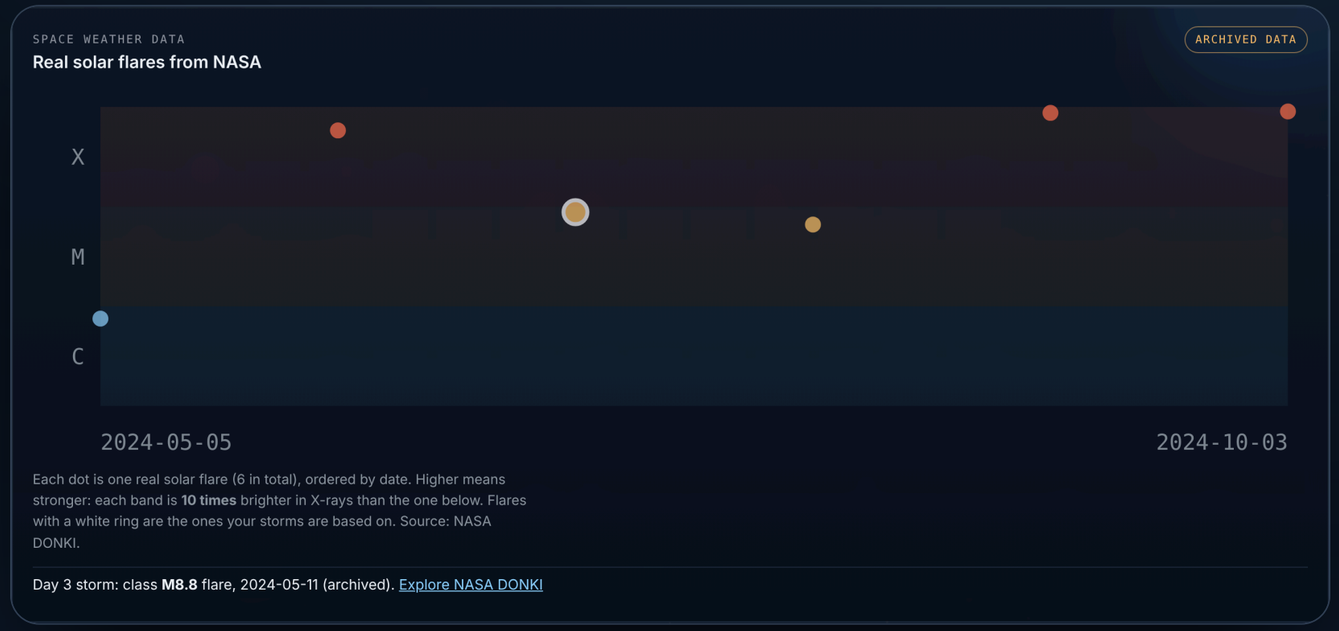
- **7-step briefing** before the game, with a practice row.
- **First-day tour** that points at each part of the screen.
- **Daily advice** that checks your plan, like "the air runs out in 3 days".
- An encouraging **debrief** with a tip for next time.



NASA DATA

Every storm is a real solar flare from NASA DONKI

- The game loads the last 30 days of flares from NASA's **DONKI** space weather database.
- Stronger flares (C, M, X class) make bigger storms.
- Players see every flare on a chart and can open its NASA record.
- Offline? It falls back to real 2024 flares, clearly labelled as archived.



SCIENCE

Mission Lab: test it like a scientist

Question, hypothesis, experiment, results, conclusion. Players predict which plan works best, then run a **fair test**: every plan faces the same 500 missions.

~10% "Steady" (same plan every day) survives

~8% "Science first" survives, but scores the most

~100% "Watch the weather" survives, with a higher score

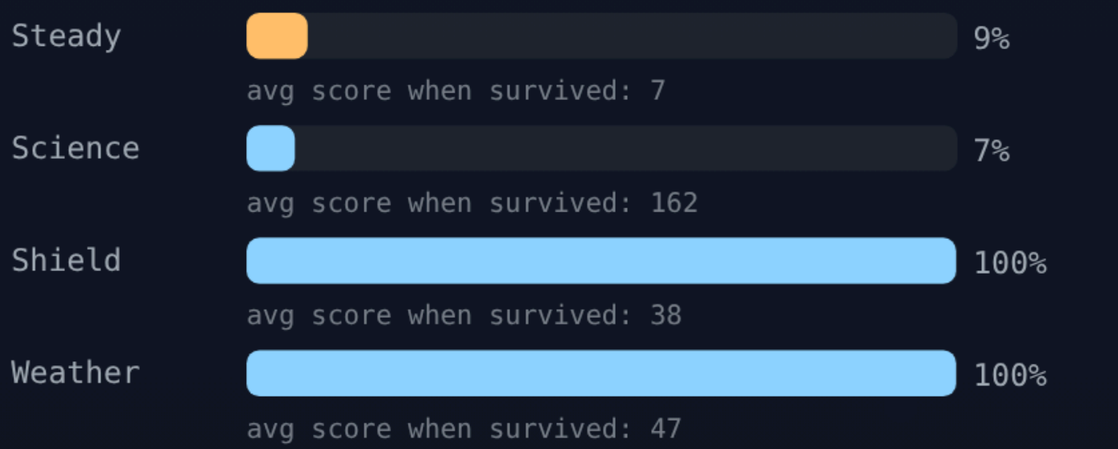
The surprise teaches why testing beats guessing.

3. EXPERIMENT

The computer plays **500 missions with each plan**. It's a fair test: every plan faces exactly the same 500 missions, with storms from the same real NASA flares. Only the plan changes.

Run 2,000 missions

4. RESULTS



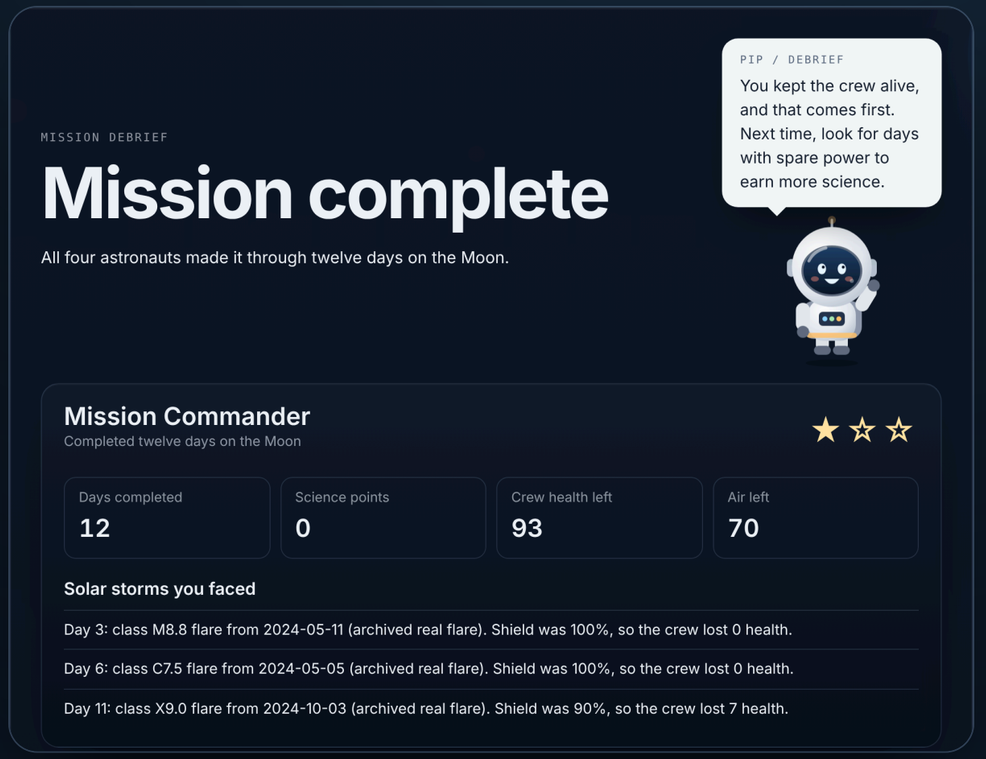
Bars: % of 500 missions the crew survived. Your prediction is amber.

IMPACT

Lessons that matter back on Earth

- **Solar power and batteries** for off-grid homes and schools face the same planning problem as the lunar night.
- **Space weather** can disturb satellites, GPS and power grids on Earth.
- An **international crew** (India, Canada, Japan, Brazil) shows space is shared.
- **Next:** Tamil and Hindi versions for local schools, and a Mars mode.

Play: itzzmeeneha.github.io/moon-base-mission



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